Grounded Distractor Probing: A Diagnostic Protocol for Hard-Distractor Sensitivity in VLM Multiple-Choice Evaluation

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Abstract

Multiple-choice evaluation for vision-language models (VLMs) increasingly relies on stronger distractors, but recent work has mostly emphasized distractor construction or generic option bias. We introduce Grounded Distractor Probing (GDP), a matched original/permutation/hard diagnostic protocol that probes how models respond to same-type, visually plausible near-miss distractors under a controlled intervention. GDP is instantiated on GDP-Slice, a manually curated 98-item diagnostic slice, and additionally characterized on an overlap-inclusive 198-item GDP-v2-lite silver set. Across four VLMs, we find that evidence-first prompting is not a universal improvement: under the same hard-distractor probe, some models improve hard-condition accuracy or reduce distractor selection, while others show lower hard-condition accuracy or increased distractor selection. We therefore argue that MCQ-based VLM evaluation should report distractor sensitivity together with model-dependent intervention response, rather than aggregate accuracy alone.

CCS Concepts

• **Computing methodologies** → **Computer vision**; **Natural language processing**; • **Information systems** → *Question answering*.

Keywords

Vision-language models, Multiple-choice question answering, Distractor analysis

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1 Introduction

When a vision-language model selects the correct option in a multiple-choice question, does the answer truly reflect fine-grained visual grounding, or could it arise from coarse cues that bypass careful image inspection?

Recent VLM evaluation has increasingly adopted multiple-choice formats, with benchmark efforts such as VMCBench and MMBench converting or standardizing diverse visual question answering tasks into unified MCQ settings [17, 30]. More broadly, the MCQA evaluation paradigm has documented format-level limitations [3] and prior work has shown that MCQ-style evaluation can be affected by generic option-position or token-level selection bias [1, 20, 31, 32], while multiple-choice VQA benchmarks may contain shortcut-enabling dataset biases [25] or admit blind solutions that bypass the image entirely [4, 11]. In contrast to broad benchmark efforts such as MMMU [27] and its harder successor MMMU-Pro [28], our goal is not to estimate benchmark-scale prevalence, but to probe a narrower failure mode: same-type, visually plausible distractors that require fine-grained visual evidence to reject. Our work focuses on this setting, which we refer to as hard-distractor sensitivity, and studies it through a controlled diagnostic protocol.

Recent work has strengthened MCQ-based VLM evaluation through automated conversion of open-ended VQA into challenging MCQ at scale [30] and robust benchmark design. Complementary probing benchmarks show that controlled foils and fluent hard negatives can reveal failures hidden by aggregate accuracy [9, 21, 22]. Robustness work further shows that distraction effects and prompt-based mitigation can vary across models and tasks [15]. Related benchmarks such as visual-language-prior probes target visual reasoning against language priors, but use a different task structure from same-type MCQ distractors [18].

Our contribution is diagnostic rather than benchmark-oriented, complementing diagnostic suites that target other VLM failure modes such as hallucination and visual illusion [8]. We do not claim to discover hard distractors as a new phenomenon, nor do we aim to construct a benchmark at scale. Instead, we use a manually curated diagnostic slice and a matched original/permutation/hard protocol to ask a more specific question: when the same evidence-first intervention is applied under same-type near-miss distractors, do models exhibit similar or divergent response patterns?

Under this framing, evidence-first is used as a controlled probe of model-dependent intervention response rather than as a new prompting method.

Contributions.

- We introduce GDP, a matched three-condition diagnostic protocol (*original*, *permutation-control*, *hard-distractor*) for probing hard-distractor-specific behavior relative to generic order effects.

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- We construct GDP-Slice, a small-but-curated diagnostic slice with manually curated same-type near-miss distractors, report a 40-item rule-based silver analysis, and use GDP-v2-lite as a relaxed 198-item supplementary check.
- We show that the same evidence-first intervention is not a universal remedy under hard distractors: it exhibits model-dependent effects, improving hard-condition behavior for some VLMs while yielding non-positive effects for others.

2 Grounded Distractor Probing

2.1 Task Setup

GDP consists of three evaluation conditions for each question-image pair: **original**, which keeps the original multiple-choice item unchanged; **permutation-control**, which preserves option meanings but reorders their positions; and **hard-distractor**, which replaces one incorrect option with a same-type, visually plausible distractor designed to require fine-grained visual evidence for rejection.

We compare **direct** answering with **evidence-first** answering. In the latter setting, the model is instructed to state the image cue it relies on before selecting an option, following vision-language chain-of-thought prompting [5, 29] and related visual-evidence prompting [13].

We define a hard distractor as an incorrect option that (i) preserves the same answer type as the gold answer, (ii) remains visually plausible under a coarse reading of the image, and (iii) is manually curated so that rejecting it requires localized fine-grained visual evidence rather than language priors alone. Under this definition, hard distractors are not merely harder wrong answers; rather, they are near-miss alternatives designed to expose failures of precise visual grounding beyond coarse answer-type matching or generic order effects.

Operational criteria for hard distractors. We manually curate GDP-Slice distractors using the following operational criteria: (i) **same-type matching**: the distractor matches the gold answer's answer-type family and, when applicable, its surface format (e.g., number, decimal precision, percentage, or time format); (ii) **near-miss plausibility**: it remains visually plausible under a coarse reading of the image; (iii) **fine-grained rejection**: rejecting it requires a localized or fine-grained visual cue rather than a coarse semantic mismatch; and (iv) **non-trivial eliminability**: it should not be removable primarily through language priors alone. Original distractors in the source MCQ items need not satisfy these constraints.

These criteria are operational construction rules; the 40-item diagnostics reported below are automatic proxies rather than human-validated labels.

The answer-type families used in GDP-Slice are *number*, *decimal*, *count*, *percentage*, *time*, and *other*, where the last category includes code, phone-number, spatial-relation, and traffic-sign items.

Figure 1 illustrates representative GDP-Slice cases where the hard distractor preserves the answer type while requiring a localized visual cue for rejection.

Table 1: GDP-Slice profile and rule-based silver analysis. Rates are computed over available automatic proxy outputs.

Composition	
Initial curation batch	43
Expansion batch	55
Total items	98
number	39
decimal	26
count	13
percentage	11
time	3
other	6
Rule-based silver analysis	
Answer-type consistency proxy	40/40 (1.000)
Heuristic plausibility proxy	37/38 (0.974)
Model-based distinguishability proxy	29/35 (0.829)

2.2 GDP-Slice Construction and Diagnostics

Source items and MCQ format. GDP-Slice is derived from the development split of VMCBench (`VMCBench_DEV.tsv`) [30]. We select 98 items from the TextVQA, InfoVQA, DocVQA, ChartQA, and RealWorldQA portions of this split and manually curate one same-type near-miss hard distractor for each item. Each item pairs one image with one question and four textual answer options (A–D), with exactly one gold answer.

For the hard-distractor condition, we preserve the gold answer and replace exactly one original incorrect option with the manually curated same-type near-miss distractor. In the primary fixed-order construction, the inserted distractor occupies the slot of the replaced option; its position is therefore item-specific rather than globally shuffled (A: 2 items, B: 1, C: 28, and D: 67). The matched permutation-control uses the reversed option order (A,B,C,D → D,C,B,A), while five additional permutation orders are used only in the auxiliary robustness analysis.

Construction and proxy diagnostics. Candidates were prioritized using answer-type matching, surface-format similarity, and near-miss heuristics, then manually curated under the operational criteria above. The final slice was assembled in two curation batches: 43 initial items and 55 expansion items (98 total; Table 1). Its 92 numeric/OCR-heavy items dominate the six remaining code, phone-number, spatial-relation, and traffic-sign items, so we make no standalone claims for the latter subset.

For a 40-item subset, we compute rule-based silver proxies for answer-type consistency, plausibility, and distinguishability. These proxies are derived from construction metadata, numerical-distance and surface-format heuristics, and precomputed model outcomes, respectively. Missing proxy values yield denominators of 40, 38, and 35; these values are not human annotations or an inter-annotator validation study.

3 Experimental Setup

3.1 Models

We evaluate four open-source VLMs with different training lineages: LLaVA-1.5-13B [14], LLaVA-OneVision-7B [12], Qwen2.5-VL-7B [2], and InternVL2.5-8B [6]. Open-source models enable controlled local

inference and within-model comparisons under fixed generation settings.

3.2 Inference Protocol

Within each evaluation condition, direct and evidence-first runs use the same image, question, and condition-specific option set. Unless otherwise noted, decoding settings are identical across the two prompting modes.

Direct prompt. In the direct setting, each model receives the image, question, and answer options and produces a direct selection. In practice, the returned answer is parsed from the selected option marker; depending on the model, this may appear either as a bare option label (e.g., `A`) or as an option label with its text (e.g., `A. 12:57`).

Evidence-first prompt. In the evidence-first setting, each model receives the same inputs but is instructed to first produce a short evidence statement and then a final answer. The resulting outputs follow the pattern `EVIDENCE: <brief visual cue>` and `FINAL_ANSWER: <selected option>`.

Decoding and parsing. For evidence-first outputs, we recover the final prediction from the `FINAL_ANSWER` field, mapping option labels or option strings to the corresponding choice. We record tag compliance for `EVIDENCE` and `FINAL_ANSWER` to separate intervention effects from formatting failures. Outputs without a recoverable option label are counted as incorrect. We parse answers from the generated text rather than from first-token probabilities, since these can diverge substantially in instruction-tuned models [19, 24].

3.3 Metrics

Let $A_{c,p}$ denote accuracy under condition $c \in \{\text{orig}, \text{perm}, \text{hard}\}$ and prompting mode $p \in \{\text{D}, \text{EF}\}$. In the main diagnostic tables, we focus on permutation-control and hard-distractor accuracies, since HSIG is defined relative to permutation-control gains. We also report the hard-distractor selection rate $\text{HDsel}_p = n_p^{\text{HD}}/N_{\text{hard}}$, where n_p^{HD} is the number of hard-condition items for which the inserted hard distractor is selected.

To quantify hard-specific intervention effects beyond generic prompt gains, we define

$$\text{HSIG} = \left(A_{\text{hard,EF}} - A_{\text{hard,D}}\right) - \left(A_{\text{perm,EF}} - A_{\text{perm,D}}\right).$$

We also report $\Delta\text{HDsel} = \text{HDsel}_{\text{EF}} - \text{HDsel}_{\text{D}}$.

Positive HSIG denotes greater improvement on hard than permutation items, whereas negative HSIG denotes hard-specific degradation; negative ΔHDsel is favorable.

We assess hard-condition direct-versus-evidence-first differences using McNemar tests and 95% bootstrap confidence intervals, with Bonferroni correction across the four model-wise comparisons.

4 Results

4.1 Main Results

Figure 2 and Table 2 show that evidence-first produces model-dependent intervention effects under hard distractors. LLaVA-1.5-13B and LLaVA-OneVision-7B show positive hard-condition gains (+10.2 and +5.1 points), whereas Qwen2.5-VL-7B and InternVL2.5-8B show negative gains (-6.1 and -3.1 points). Using permutation-control gains as a reference, HSIG remains positive for LLaVA-1.5-13B (+8.2 points) and LLaVA-OneVision-7B (+5.1 points), but negative for Qwen2.5-VL-7B (-4.1 points) and InternVL2.5-8B (-3.1 points). LLaVA-OneVision-7B also reduces hard-distractor selection (ΔHDsel = -0.041), whereas Qwen2.5-VL-7B increases it (ΔHDsel = +0.041).

For Qwen2.5-VL-7B, the negative effect is not attributable to label recovery failure: final prediction labels were recovered for all items under both prompting settings. For InternVL2.5-8B, malformed evidence-first outputs were concentrated in a single item, and excluding that item did not materially change the non-positive hard-condition conclusion.

The strongest positive signal appears in LLaVA-1.5-13B, whose hard-condition gain is +10.2 points (McNemar $p = 0.006$; 95% bootstrap CI [0.041, 0.174]). By contrast, Qwen2.5-VL-7B shows a directionally negative hard-condition gain of -6.1 points (McNemar $p = 0.070$; 95% bootstrap CI [-0.122, -0.010]). We report uncorrected McNemar p-values for transparency; under Bonferroni correction for four model-wise tests ($\alpha = 0.0125$), only LLaVA-1.5-13B remains significant, while the Qwen2.5-VL-7B result should be interpreted as directionally negative with a fully negative bootstrap CI but a non-significant McNemar test. LLaVA-OneVision-7B shows a weaker positive pattern (95% bootstrap CI [-0.010, 0.122]), whereas InternVL2.5-8B remains non-positive overall (95% bootstrap CI [-0.082, 0.010]).

Table 2 uses the reversed option order (A,B,C,D → D,C,B,A) as the matched permutation-control. As an auxiliary check, we ran five additional permutation-control orders on the strict 98-item GDP-Slice and recomputed HSIG using the average permutation gain over the existing and added permutation controls; the signs are preserved (LLaVA-1.5: +0.119; LLaVA-OneVision: +0.039; Qwen2.5-VL: -0.044; InternVL2.5: -0.029), suggesting that the pattern is not driven by a single option order.

4.2 Ablation and Robustness Analysis

Image-blind sanity check. Removing the image in a question-only (qonly) control on GDP-Slice lowers hard-condition accuracy for Qwen2.5-VL-7B, LLaVA-OneVision-7B, and InternVL2.5-8B, with increased hard-distractor selection. LLaVA-1.5-13B drops less because its image-conditioned hard accuracy is already low, suggesting that the same-type near-miss distractors are not generally removable from question-option text alone, consistent with concerns about

Table 2: Diagnostic results on GDP-Slice. P = permutation-control, H = hard-distractor, D = direct, EF = evidence-first. HSIG estimates hard-specific intervention gain relative to generic prompt effects.

Model	P-D	H-D	P-EF	H-EF	HSIG	ΔHDsel
LLaVA-1.5-13B	0.296	0.337	0.316	0.439	+0.082	+0.000
LLaVA-OneVision-7B	0.796	0.755	0.796	0.806	+0.051	-0.041
Qwen2.5-VL-7B	0.847	0.857	0.827	0.796	-0.041	+0.041
InternVL2.5-8B	0.908	0.888	0.908	0.857	-0.031	+0.000

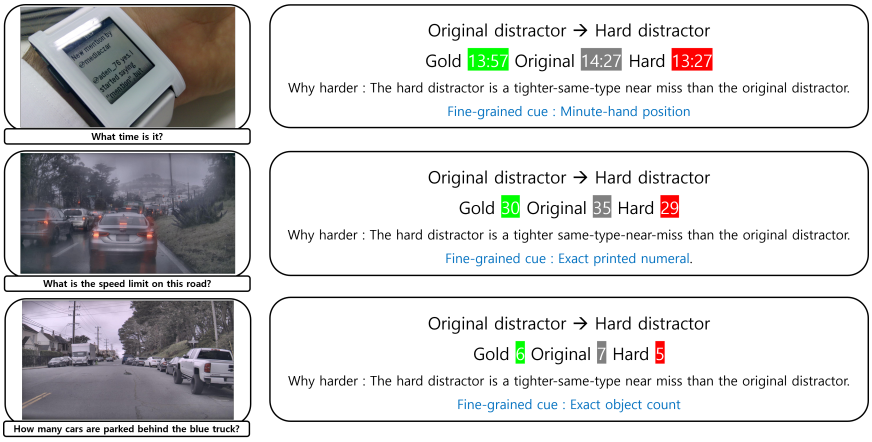


Figure 1: Representative GDP-Slice cases showing same-type near-miss hard distractors. The hard distractor preserves the answer type of the gold answer and is designed to require a localized fine-grained visual cue for rejection.

image-blind shortcuts in MCQ evaluation [7, 11]. The analogous qonly check on GDP-v2-lite is summarized in Table 3.

Answer-first control. As a targeted follow-up on two opposite-response models, we ran an answer-first control on Qwen2.5-VL-7B and LLaVA-OneVision-7B under permutation and hard conditions. For LLaVA-OneVision-7B, answer-first preserves the hard-specific accuracy signal ($HSIG_{AF}$ = +0.051) but weakens the selection-side gain ($\Delta HDsel_{AF}$ = -0.031 vs. -0.041). For Qwen2.5-VL-7B, answer-first returns accuracy to the direct baseline ($HSIG_{AF}$ = 0.000) and reduces hard-distractor confusion relative to evidence-first. Thus, Qwen's negative evidence-first response is order-sensitive, whereas OneVision retains the accuracy-side signal but weakens on selection-side behavior when evidence follows the answer.

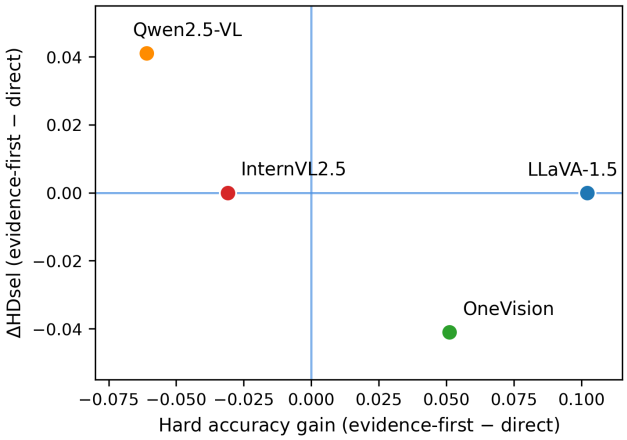


Figure 2: Model-dependent effects of evidence-first prompting on GDP-Slice. The x-axis shows hard-condition accuracy gain (EF - direct), and the y-axis shows ΔHDsel. Evidence-first helps some models but reduces hard-condition accuracy or increases distractor selection for others.

Table 3: Relaxed GDP-v2-lite sanity check (198 items). Qonly-H denotes hard-condition direct accuracy with the image removed; Gap is H-D minus Qonly-H.

Model	HSIG	H-D	Qonly-H	Gap
LLaVA-1.5-13B	+0.000	0.328	0.247	+0.081
LLaVA-OneVision-7B	+0.030	0.646	0.308	+0.338
Qwen2.5-VL-7B	-0.040	0.702	0.247	+0.455
InternVL2.5-8B	+0.020	0.692	0.354	+0.338

Relaxed v2-lite sanity check. We also evaluate GDP-v2-lite, a relaxed silver-level slice of 198 items with automatically proposed and filtered hard distractors. Although still numeric and OCR-heavy, it provides a larger-scale sanity check on whether the diagnostic pattern is confined to the strict 98-item GDP-Slice. As shown in Table 3, evidence-first remains non-universal: LLaVA-OneVision has a positive HSIG point estimate (+0.030), whereas Qwen2.5-VL has a negative one (-0.040). In the relaxed slice, the LLaVA-1.5 hard-specific gain is attenuated and InternVL2.5 shifts to a weakly positive mixed profile; accordingly, we treat GDP-v2-lite as a scale sanity check rather than a replacement for the strict GDP-Slice. In qonly controls, hard accuracy is lower for all models, with large gaps for LLaVA-OneVision, Qwen2.5-VL, and InternVL2.5.

Hard-distractor-specific rescue and damage. We decompose hard-condition transitions into rescue (direct wrong, EF correct) and damage (direct correct, EF wrong), further marking rescue-from-HD and damage-to-HD cases. Table 4 shows that LLaVA-OneVision-7B has several rescue-from-HD cases with reduced hard-distractor selection, whereas Qwen2.5-VL-7B shows more damage than rescue, including damage-to-HD cases. This supports interpreting evidence-first as an intervention that can either correct or reinforce hard-distractor confusion.

Table 4: Hard-distractor-specific rescue and damage on GDP-Slice.

Model	Rescue	Rescue-from-HD	Damage	Damage-to-HD
LLaVA-1.5-13B	11	1	1	1
LLaVA-OneVision-7B	8	4	3	2
Qwen2.5-VL-7B	1	0	7	3
InternVL2.5-8B	1	0	4	1

Evidence-target alignment audit. Among 57 inspected traces (21 rescue; 36 damage or stable-wrong), 14/21 rescue traces are gold-supporting, whereas 33/36 damage or stable-wrong traces are wrong-option-supporting, generic/empty, or target-unclear, consistent with rationale–answer alignment analysis [26]. This audit measures evidence–answer target alignment, not visual correctness or localization.

5 Discussion and Limitations

Taken together, the strict 98-item GDP-Slice and relaxed GDP-v2-lite show that evidence-first prompting is not a uniform hard-distractor remedy [10, 16, 23]. The strict slice shows a sharp model-dependent profile: both LLaVA variants show positive hard-condition point estimates, Qwen2.5-VL shows a negative hard-specific response with increased hard-distractor selection, and InternVL2.5 is non-positive. In GDP-v2-lite, the profiles become less sharp: Qwen2.5-VL remains negative in HSIG across both slices (HSIG ≈ -0.04), LLaVA-1.5's hard-specific gain becomes generic, and InternVL2.5 shifts to a weakly positive mixed profile. We therefore use evidence-first as a diagnostic intervention and report intervention profiles rather than model or prompt rankings. Similar aggregate accuracy changes can arise from different answer-transition mechanisms, as the rescue/damage analysis shows.

The study has several limitations. The strict 98-item GDP-Slice is small and manually curated. The larger GDP-v2-lite provides a limited scale sanity check but remains numeric/OCR-heavy, silver-level, and not benchmark-scale. The main GDP-Slice HSIG table uses one reversed-order permutation control, although an auxiliary check with five additional orders preserves the qualitative signs. We evaluate four open-source VLMs, and our evidence audit measures answer-target alignment rather than visual faithfulness or localization correctness.

GDP inserts exactly one hard distractor per item; whether two or three simultaneous near-miss alternatives alter the observed intervention profiles remains untested. The results should therefore be read as controlled evidence for a specific failure mode rather than as a prevalence estimate. Future work should use multi-permutation averaging, broader answer-type coverage, more model families, direct evidence-quality annotation, and multi-distractor settings.

Accordingly, these artifacts are intended as reference diagnostic slices and scripts for applying GDP to other MCQ settings, not as a leaderboard benchmark. Code, prompt templates, metric scripts, and GDP-Slice/GDP-v2-lite annotations will be released upon publication, subject to source-dataset licenses.

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GenAI Usage Disclosure

GenAI tools were used solely for grammar and wording refinement in this manuscript. All experimental design, data curation, code implementation, statistical analysis, and interpretation were performed and verified by the authors.

References

[1] Md. Atabuzzaman, Ali Asgarov, and Chris Thomas. 2025. Benchmarking and Mitigating MCQA Selection Bias of Large Vision-Language Models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. 33548–33562.

[2] Shuai Bai et al. 2025. Qwen2.5-VL Technical Report. *arXiv preprint arXiv:2502.13923* (2025).

[3] Nishant Balepur, Rachel Rudinger, and Jordan Lee Boyd-Graber. 2025. Which of these best describes multiple choice evaluation with llms? a) forced b) flawed c) fixable d) all of the above. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 3394–3418.

[4] Lin Chen, Jinsong Li, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Zehui Chen, Haodong Duan, Jiaqi Wang, Yu Qiao, Dahua Lin, et al. 2024. Are we on the right way for evaluating large vision-language models? *Advances in Neural Information Processing Systems* 37 (2024), 27056–27087.

[5] Yangyi Chen, Karan Sikka, Michael Cogswell, Heng Ji, and Ajay Divakaran. 2024. Measuring and improving chain-of-thought reasoning in vision-language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. 192–210.

[6] Zhe Chen, Weiyun Wang, Yue Cao, Yangzhou Liu, Zhangwei Gao, Erfei Cui, Jinguo Zhu, Shenglong Ye, Hao Tian, Zhaoyang Liu, et al. 2024. Expanding performance boundaries of open-source multimodal models with model, data, and test-time scaling. *arXiv preprint arXiv:2412.05271* (2024).

[7] Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. 2017. Making the v in vqa matter: Elevating the role of image understanding in visual question answering. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 6904–6913.

[8] Tianrui Guan, Fuxiao Liu, Xiyang Wu, Ruiqi Xian, Zongxia Li, Xiaoyu Liu, Xijun Wang, Lichang Chen, Furong Huang, Yaser Yacoob, et al. 2024. Hallusionbench: an advanced diagnostic suite for entangled language hallucination and visual illusion in large vision-language models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 14375–14385.

[9] Cheng-Yu Hsieh, Jieyu Zhang, Zixian Ma, Aniruddha Kembhavi, and Ranjay Krishna. 2023. Sugarcrepe: Fixing hackable benchmarks for vision-language compositionality. *Advances in neural information processing systems* 36 (2023), 31096–31116.

[10] Dongzhi Jiang, Renrui Zhang, Ziyu Guo, Yanwei Li, Yu Qi, Xinyan Chen, Liuhui Wang, Jianhan Jin, Claire Guo, Shen Yan, Bo Zhang, Chaoyou Fu, Peng Gao, and Hongsheng Li. 2025. MME-CoT: Benchmarking Chain-of-Thought in Large Multimodal Models for Reasoning Quality, Robustness, and Efficiency. In *Proceedings of the 42nd International Conference on Machine Learning*, Vol. 267. PMLR, 27793–27830.

[11] Baiqi Li, Zhiqiu Lin, Wenxuan Peng, Jean de Dieu Nyandwi, Daniel Jiang, Zixian Ma, Simran Khanuja, Ranjay Krishna, Graham Neubig, and Deva Ramanan. 2024. Naturalbench: Evaluating vision-language models on natural adversarial samples. *Advances in Neural Information Processing Systems* 37 (2024), 17044–17068.

[12] Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Peiyuan Zhang, Yanwei Li, Ziwei Liu, et al. 2024. Llava-onevision: Easy visual task transfer. *arXiv preprint arXiv:2408.03326* (2024).

[13] Wei Li, Zhen Huang, Houqiang Li, Le Lu, Yang Lu, Xinmei Tian, Xu Shen, and Jieping Ye. 2025. Visual Evidence Prompting Mitigates Hallucinations in Large Vision-Language Models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 4048–4080.

[14] Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. 2024. Improved baselines with visual instruction tuning. In *Proceedings of the IEEE/CVF conference on*

computer vision and pattern recognition. 26296–26306.

[15] Ming Liu, Hao Chen, Jindong Wang, Liwen Wang, Jingchen Sun, and Wensheng Zhang. 2026. Is Your (Reasoning) Multimodal Language Model Vulnerable Toward Distractions?. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 40. 32087–32095.

[16] Ryan Liu, Jiayi Geng, Addison J Wu, Ilia Sucholutsky, Tania Lombrozo, and Thomas L Griffiths. 2024. Mind your step (by step): Chain-of-thought can reduce performance on tasks where thinking makes humans worse. *arXiv preprint arXiv:2410.21333* (2024).

[17] Yuan Liu, Haodong Duan, Yuanhan Zhang, Bo Li, Songyang Zhang, Wangbo Zhao, Yike Yuan, Jiaqi Wang, Conghui He, Ziwei Liu, et al. 2024. Mmbench: Is your multi-modal model an all-around player?. In *European conference on computer vision*. Springer, 216–233.

[18] Tiange Luo, Ang Cao, Gunhee Lee, Justin Johnson, and Honglak Lee. 2025. Probing Visual Language Priors in VLMs. In *Proceedings of the 42nd International Conference on Machine Learning*, Vol. 267. PMLR, 41120–41156.

[19] Francesco Maria Molfese, Luca Moroni, Luca Gioffré, Alessandro Scirè, Simone Conia, and Roberto Navigli. 2025. Right answer, wrong score: Uncovering the inconsistencies of LLM evaluation in multiple-choice question answering. In *Findings of the Association for Computational Linguistics: ACL 2025*. 18477–18494.

[20] Pouya Pezeshkpour and Estevam Hruschka. 2024. Large language models sensitivity to the order of options in multiple-choice questions. In *Findings of the Association for Computational Linguistics: NAACL 2024*. 2006–2017.

[21] Tristan Thrush, Ryan Jiang, Max Bartolo, Amanpreet Singh, Adina Williams, Douwe Kiela, and Candace Ross. 2022. Winoground: Probing vision and language models for visio-linguistic compositionality. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 5238–5248.

[22] Shengbang Tong, Zhuang Liu, Yuexiang Zhai, Yi Ma, Yann LeCun, and Saining Xie. 2024. Eyes wide shut? exploring the visual shortcomings of multimodal llms. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 9568–9578.

[23] Miles Turpin, Julian Michael, Ethan Perez, and Samuel Bowman. 2023. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. *Advances in Neural Information Processing Systems* 36 (2023), 74952–74965.

[24] Xinpeng Wang, Bolei Ma, Chengzhi Hu, Leon Weber-Genzel, Paul Röttger, Frauke Kreuter, Dirk Hovy, and Barbara Plank. 2024. "My answer is C": First-token probabilities do not match text answers in instruction-tuned language models. In *Findings of the Association for Computational Linguistics: ACL 2024*. 7407–7416.

[25] Zhecan Wang, Long Chen, Haoxuan You, Keyang Xu, Yicheng He, Wenhao Li, Noel Codella, Kai-Wei Chang, and Shih-Fu Chang. 2023. Dataset bias mitigation in multiple-choice visual question answering and beyond. In *Findings of the Association for Computational Linguistics: EMNLP 2023*. 8598–8617.

[26] Yuanchen Wu, Ke Yan, Shouhong Ding, Ziyin Zhou, and Xiaoqiang Li. 2025. Towards Rationale-Answer Alignment of LVLMs via Self-Rationale Calibration. In *Proceedings of the 42nd International Conference on Machine Learning*, Vol. 267. PMLR, 67938–67958.

[27] Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, et al. 2024. Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 9556–9567.

[28] Xiang Yue, Tianyu Zheng, Yuansheng Ni, Yubo Wang, Kai Zhang, Shengbang Tong, Yuxuan Sun, Botao Yu, Ge Zhang, Huan Sun, et al. 2025. Mmmu-pro: A more robust multi-discipline multimodal understanding benchmark. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 15134–15186.

[29] Ruohong Zhang, Bowen Zhang, Yanghao Li, Haotian Zhang, Zhiqing Sun, Zhe Gan, Yinfei Yang, Ruoming Pang, and Yiming Yang. 2025. Improve vision language model chain-of-thought reasoning. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 1631–1662.

[30] Yuhui Zhang, Yuchang Su, Yiming Liu, Xiaohan Wang, James Burgess, Elaine Sui, Chenyu Wang, Josiah Aklilu, Alejandro Lozano, Anjiang Wei, et al. 2025. Automated generation of challenging multiple-choice questions for vision language model evaluation. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 29580–29590.

[31] Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. 2024. Large language models are not robust multiple choice selectors. In *International Conference on Learning Representations*.

[32] Yongshuo Zong, Tingyang Yu, Ruchika Chavhan, Bingchen Zhao, and Timothy Hospedales. 2023. Fool your (vision and) language model with embarrassingly simple permutations. *arXiv preprint arXiv:2310.01651* (2023).